

Akshay Dahiya

Protocol Engineer with 8+ years building blockchain infrastructure, smart contracts, and distributed systems.

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TECHNICAL SKILLS

Protocol & Tokenomics

Protocol Design, Game Theory, Consensus, Staking/Slashing, Tax Curves, Airdrops, Sybil Resistance, Incentivized Testnets, Emission Schedules, Delegation, Incentive Alignment

Smart Contracts

Solidity, Upgradeable Proxies, EIP-712, Gasless Relayers, VRF, Oracles, ERC20/721/1155, Audited Code, Batching, SBTs

Decentralized Systems

Snapshotter Networks, Sequencers, IPFS/Data Availability, Prediction Markets, Points Systems, Agentic AI

Full Stack & DevOps

Python, Node.js, Rust, Linux/Shell, Docker, K8s, CI/CD, AWS, Redis, React, Next.js, High-Throughput APIs

Web3 Tooling

Hardhat, Foundry, Ethers.js, Web3.py, IPFS, Data Indexing, Chainlink

AI & Machine Learning

PyTorch, TensorFlow, LLMs, Reinforcement Learning (PPO), Computer Vision, Agentic AI

Leadership & Other

Technical Writing, Team Leadership, Public Speaking

KEY HIGHLIGHTS

- **800 ETH raised** and **5,300+ SBT mints** via protocol staking contracts
- **1M+ gasless transactions** sponsored through custom EIP-712 relayers
- **50+ audited smart contracts** deployed across major EVM networks
- **3M+ daily requests** served by high-scale API gateway infrastructure
- **2 ACM publications**; AI research featured in Forbes
- **2,000+ projects reviewed**, mentoring 500+ aspiring engineers

PROFESSIONAL EXPERIENCE

Lead Protocol Engineer & DeFi Researcher

Dec 2022 - Present

POWERLOOM PROTOCOL

Remote

- Co-designed core protocol evolution from V0 → V1 → V2 → DSV architecture
- Engineered 'Snapshotter' node registry contracts, securing **5,300+ nodes** and **800 ETH** raised
- Designed **\$POWER tokenomics**, staking modules, and airdrop mechanics on Arbitrum Nitro
- Built high-throughput **EIP-712 relayers** ensuring gasless txs for **1M+ interactions**
- Developed **Onchain Points** and Generative Prediction Markets using live protocol data
- Led engineering for **3 incentivized testnets** with CoinList, Polygon zkEVM, and Socket
- Single-handedly shipped **BDS (Blockchain Data Services)**: decentralized real-time Uniswap data market with IPFS proofs

Senior Software Engineer

Mar 2020 - Dec 2022

HYPERSONIX INC.

Remote

- Engineered high-availability **API Gateway** scaling to **3M+ daily requests** for enterprise data
- Optimizing API latency by **80%** using Brotli compression and query tuning
- Implemented enterprise auth with Auth0 (SSO, OAuth 2.0) to secure sensitive endpoints
- Deployed comprehensive ELK observability stack for real-time monitoring

Blockchain Developer

Dec 2020 - Nov 2022

FREELANCE

Remote

- Delivered **50+ audited smart contracts** across EVM chains, prioritizing gas optimization
- Architected complex tokens (ERC-20/721/1155) with deflationary and dynamic reward logic
- Built DeFi protocols including MEV-resistant vaults and automated referral systems
- Developed automated trading infrastructure using Web3.py and Solidity for NFT and token markets
- Engineered Play-to-Earn ecosystems integrated with Chainlink VRF and cross-chain bridges

Graduate Programmer

FIDELITY INTERNATIONAL

- Built real-time survey platform using WebSockets, Django, Redis, and D3.js
- Developed React-based analytical tools for behavioral finance portfolio managers

Jun 2018 - Nov 2018

Bangalore, India

Technical Mentor & Project Reviewer

UDACITY

- Reviewed **2,000+ technical projects** across Full Stack and AI nanodegrees
- Mentored **500+ students** globally through code reviews and debugging sessions

Feb 2017 - Mar 2020

Remote

Open Source Contributor & Admin

GOOGLE SUMMER OF CODE — HYDRA ECOSYSTEM

- Core contributor to Hydra Ecosystem — framework for smart, self-describing Web APIs
- Served as GSoC Admin/Mentor (2018–2021), guiding **10+ contributors** on open-source tooling

Feb 2018 - Aug 2021

Remote

EDUCATION

Master of Science in Computer Science — Machine Learning Specialization

GEORGIA INSTITUTE OF TECHNOLOGY

- Research: Built **Examinator** (ML plagiarism detection) and **JackMarker**; published 2 ACM papers
- Coursework: Machine Learning, Reinforcement Learning, Computer Vision, AI, Graduate Algorithms

2019 - 2021

Atlanta, GA

Bachelor of Technology

DCRUST

- Comprehensive training in electronics, communication systems, and core engineering principles
- Strong foundations in signal processing, embedded systems, and technical problem-solving

2014 - 2018

India

Nanodegree Certifications

UDACITY

- **Deep Learning (2020)**: Neural networks with PyTorch — CNNs, RNNs, LSTMs, GANs
- **Computer Vision (2020, NVIDIA)**: YOLO, R-CNN, Visual SLAM, Image Captioning
- **React (2017)**: 3 full-stack apps with Redux — MyReads, Would You Rather, Mobile Flashcards
- **Full Stack (2016)**: PostgreSQL, RESTful APIs, OAuth, AWS deployment

2016 - 2020

Online

Open Courseware Foundations

MIT OCW / STANFORD / BERKELEY / EDX

- **MIT 6.006**: Core algorithms — sorting, heaps, BSTs, graph algorithms, dynamic programming
- **MIT 6.042J**: Discrete math — proofs, graph theory, probability
- **Stanford CS161**: Divide-and-conquer, randomized algorithms, NP-completeness
- **Berkeley CS188**: AI fundamentals — search, MDPs, reinforcement learning, Q-learning

2015 - 2019

Online

PUBLICATIONS

1. Dahiya, A., Graziano, R., Irish, I., Starner, T.

JackMarker with GitDown: A Framework to Counter Plagiarism at Scale

ACM Learning @ Scale 2021. DOI: 10.1145/3430895.3460982

2. Apoorv, R., Dahiya, A., Graziano, R., Starner, T.

Examinator: A Plagiarism Detection Tool for Take-Home Exams

ACM Learning @ Scale 2020. DOI: 10.1145/3386527.3406723